

Submission Summary

Conference Name

2026 IEEE 5th World Conference on Applied Intelligence and Computing

Paper ID

2760

Paper Title

Identification of Slow Learners Using Machine Learning: A Comparative Analysis of Classification Algorithms

Abstract

Abstract—We consider the problem of early identification of slow learners based on a student's performance during an ongoing term. Given that eventually one in four learners will drop out or perform extremely poorly, we investigate if it is possible to predict these students early in the term using only data available before a student has completed even 30% of their courses. We utilize 14,003 student records to analyze features such as grades, attendance, completion rate (pace), usage rate (engagement), and recency (time since last engagement). Our analysis involves six different classifiers, where top-performing models achieved near-perfect prediction accuracy.

The ablation study was the most significant finding: two specific columns—exam scores and assignment completion—were enough to perfectly replicate what all fifteen columns achieved together. Dropping those two was the only factor that significantly degraded performance. Everything else—gender, age, internet access, and stress levels—turned out to be noise. For an institution looking to deploy such a system, the message is simple: utilize the two primary metrics already present in the gradebook to train a decision tree; this provides a highly efficient and working early-warning system in milliseconds.

Index Terms—at-risk student detection, supervised classification, SHAP explanations, educational analytics, feature ablation,

ensemble methods

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Computational intelligence

Submission Files

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